

ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 432 F, SMF-28® Ultra fiber, Single-mode (OS2)

CORNING

Part Number:
432ZUC-T4122D20

Corning ALTOS® Lite gel-free, single-jacket, single-armored cables are designed for campus backbones in direct-buried installations. The loose tube design provides stable and highly reliable transmission parameters for a variety of voice, data, video and imaging applications. These cables also provide high-fiber density within a given cable diameter while allowing flexibility to suit many system configurations. The single armored construction provides additional crush and rodent protection with a high-strength ripcord under the armor for easy stripping. Gel-free means the cables are fully waterblocked using craft-friendly, water-swappable materials which make cable access simple and require no clean up. The flexible, craft-friendly buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy mid-span access. These cables have a medium density polyethylene jacket that is rugged, durable and easy to strip.

Features and Benefits

Gel-free waterblocking technology

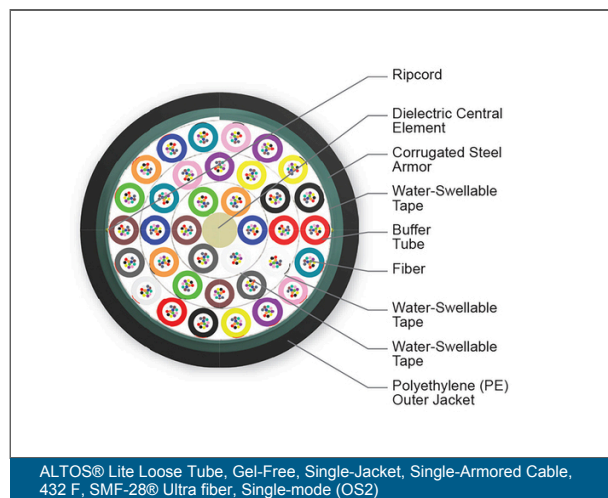
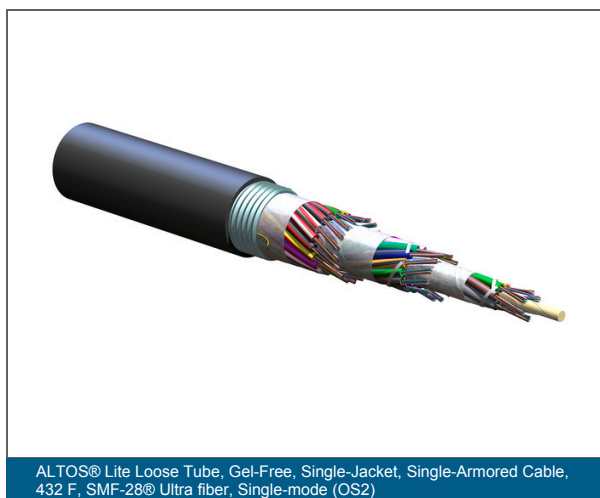
Craft-friendly cable preparation

Polyethylene jacket

Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

Corrugated steel tape armor

Provides rodent resistance for direct-buried applications



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Specifications

General Specifications

Installation Methods	Aerial, Direct Buried, Duct
Cable Type	Loose Tube
Environment	Outdoor
Product Type	Steel Tape Armor
Fiber Category	SMF-28® Ultra
Cable geometry	Round

Standards

RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Common Installations	Outdoor lashed aerial, duct and direct-buried, indoor when installed according to National Electrical Code® (NEC®) Article 770
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions

Notes	Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results.
Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-20 °C to 70 °C (-4 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design

Central Element	Dielectric
Fiber Count	432
Number of Ripcords	2
Buffer Tube Color Coding, Layer 1	Blue, Orange, Green, Brown, Slate, White

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Cable Design	
Outer Jacket Color	Black
Outer Jacket Material	Polyethylene (PE)
Buffer Tube Color Coding, Layer 3	Red*, Black*, Yellow*, Violet*, Rose*, Aqua*, Blue**, Orange**, Green**, Brown**, Slate**, White **, Red**, Black**, Yellow**, Violet**, Rose**, Aqua**
Tensile Strength Elements and/or Armoring - Layer 1	Corrugated steel tape armor
Buffer Tube Diameter	2.5 mm (0.1 in)
Number of Active Tubes	36
Number of Tube Positions	36
Tape, Layer 1	Water-swellable
Tape, Layer 2	Water-swellable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12
Color Code Standards	Telcordia

Mechanical Specifications	
Max. Tensile Strength, Long-Term	890 N (200.08 lbf)
Max. Tensile Strength, Short-Term	2700 N (606.98 lbf)
Nominal Outer Diameter	23.2 mm (0.91 in)
Min. Bend Diameter Installation	696 mm (27.4 in)
Min. Bend Diameter Operation	464 mm (18.27 in)

Optical Characteristics	
Fiber Code	Z
Performance Option Code	22
Fiber Category	OS2
Fiber Type	Single-mode (OS2) / 250 µm
Fiber Name	Bend-Improved Single-mode (OS2)
Maximum Attenuation	0.34 dB/km / 0.34 dB/km / 0.22 dB/km

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Optical Characteristics

Wavelengths	1310 nm / 1383 nm / 1550 nm
Fiber Compliance	ITU-T G.652.D and ITU-T G.657.A1

Dimensions

Cable Weight	357 kg/km (239.89 lb/1000 ft)
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